No evidence of insulin resistance in normal weight vegetarians. A case control study.

. . . . .

BACKGROUND: Diets rich in carbohydrates with a low glycemic index and with high fiber content are associated with flat post-prandial rises of blood glucose, minimal post-prandial insulin secretion and maintenance of insulin sensitivity. Protective food commodities in the prevention of cardiovascular disease, insulin resistance syndrome or diabetes are crucial components of the vegetarian diet. AIM OF THE STUDY: Insulin resistance values were assessed in relation to different nutrition. Metabolic abnormality is a predictor of age-related diseases and can be more pronounced in obese subjects. Insulin resistance values in normal weight subjects of two different nutritional habits were correlated with age. METHODS: Fasting concentrations of glucose and insulin as well as calculated values of insulin resistance IR (HOMA) were assessed in two nutritional groups of apparently healthy adult subjects (age range 19 - 64 years) with normal weight (body mass index 18.6 - 25.0 kg/m(2)): a vegetarian group (95 long-term lacto-ovo-vegetarians; duration of vegetarianism 10.2 +/- 0.5 years) and a non-vegetarian control group (107 subjects of general population on traditional western diet). Intake of energy and main nutrients (fats, saccharides, proteins) was similar in both groups. RESULTS: Glucose and insulin concentrations and IR (HOMA) values were significantly lower in vegetarians (glucose 4.47 +/- 0.05 vs. 4.71 +/- 0.07 mmol/l; insulin 4.96 +/- 0.23 vs. 7.32 +/- 0.41 mU/l; IR (HOMA) 0.99 +/- 0.05 vs. 1.59 +/- 0.10). IR (HOMA) dependence on age was only significant in subjects on a western diet. A significant increase of IR was found already in the age range 31-40 years, compared to vegetarians and it continued in later age decades. Age independent and low insulin resistance values in vegetarians are a consequence of an effective diet prevention by long-term frequent consumption of protective food. Vegetarians had a significantly higher consumption of whole grain products, pulses, products from oat and barley. CONCLUSION: The results of age independent and low values of insulin resistance document a beneficial effect of long-term vegetarian nutrition in prevention of metabolic syndrome, diabetes and cardiovascular disease.